

Wilson, Chow Sin Long

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Full-Stack Software Engineer · AWS Certified Solutions Architect – Associate

SUMMARY

Full-stack software engineer with 5 years of experience shipping **15+** React and React Native products from Figma through Node.js APIs and AWS, including App Store and Google Play releases. Cut load times by **80%** and search latency by **90%**. Now building a multi-channel **agentic** platform with durable workflows, memory, and RAG. AWS Certified Solutions Architect – Associate.

EXPERIENCE

Software Developer, Vioo Company Limited

May 2022 – Present

Key developer for concurrent client products, owning delivery from UX scoping through cloud deployment.

- Shipped and maintained **15+** React / React Native apps with high-fidelity Figma design, shared design systems, and Apple App Store / Google Play releases.
- Migrated WordPress sites to Strapi / Sanity and Meiliseach / Algolia in a **Turborepo** on **SST** without interrupting publishing; cut load times by **80%**, search latency by **90%**, and achieved PageSpeed scores above **95**.
- Built typed Node.js APIs with Hono/NestJS, OpenAPI, and tRPC on Lambda / API Gateway and **ECS Fargate**; modeled org-scoped Postgres with Drizzle / Prisma, Better Auth, OAuth 2.0 / OIDC, and RBAC.
- Owned shared **Radix** / **Tailwind** / **shadcn/ui** component and hook APIs; **Storybook**, Fumadocs, and MDX guidance eliminated duplicate UI across client apps.
- Set **Vitest**, **Playwright** / **Cypress** quality gates for shared UI and critical flows.
- Instrumented web, mobile, and API releases with **Sentry** for production crashes, device issues, and API errors.
- Adopted **tRPC** and Scalar-generated API docs, cutting integration time by **50%** and eliminating schema-mismatch errors.
- Built GitHub Actions + **SST** CI/CD with linting, type-checking, tests, and builds; cut deployment time by **50%+**.
- Mentored interns and junior developers through pairing, code reviews, and production delivery.

SELECTED PROJECTS

Multi-Channel Agentic Platform

Apr 2026 – Present

Durable agent that empowers internal tools and workflows.

- Built agent runs durable streams so clients could disconnect, reconnect, and resume without losing progress.
- Built **workflows**, **memory systems**, schedules, tools, pause/resume, retries, sandboxing, and skills in **Mastra**; **Langfuse** for observability.
- **RAG**: **structure-aware** parsing and chunking, **metadata creation**, **retrieval funneling** with **Pinecone hybrid search** (dense and sparse vectors, BM25) and reranking; **SQL agent** for experimental data analysis.
- Enforced document-level permissions with **ABAC** based on document ownership and user roles and access levels.
- Ran on **Hono**, **Better Auth**, **Postgres** with **Drizzle** ORM; **S3** for file storage; deployed to AWS ECS Fargate using **SST** as IaC.
- Built a custom **@mentions-style** input for attaching workspace files (npm: **@ocwilsonchow/editor**).

Healthcare AI Agent

July 2025 – Apr 2026

Healthcare and shopping assistant grounded in uploaded documents and Shopify store product data.

- Designed the chat and admin UX in **Figma** and owned the client-facing flows — document upload, in-chat questionnaire, product and cart actions.
- Built the backend with **Hono**, **Prisma**, **LangChain**, and **Qdrant**; hardened Next.js admin and client apps with **Better Auth**, HttpOnly cookies, CORS, and RBAC.
- Streamed **LangGraph** agent replies into the chat with the **AI SDK** over tRPC, rendering tool calls as product cards, cart actions, and questionnaires instead of plain text.
- Built an admin form builder with conditional logic for dynamic questionnaires and ran those questionnaires in-thread; modeled with **XState** and **React Hook Form** / **Zod**.
- Optimized expensive components with **memoization** and **useTransition** for heavy UI updates.
- Used **TanStack Query** with server prefetch and client-side rehydration for data lists, and **TanStack Table** with pagination, filters, and sorting; stored table state in URL search parameters so lists stay shareable and bookmarkable.
- Deployed to AWS **Lambda** (Hono API) and **OpenNext** (Nextjs) using **SST** as IaC.

High-Performance Editorial Platform

2025

Strapi, Meiliseach, and **Next.js** editorial platform in the WordPress-to-headless migration.

- Latest 100 articles statically rendered at build time, leaving the rest incrementally rendered on demand. Strapi **GraphQL** queries tagged properly with **Next.js** cache tags; revalidated through webhook triggered by content update events.
- Shipped instant search feature using Meilisearch with full-text search and fuzzy matching; result hits in < **50ms**.
- PageSpeed Insights average scores >**90**; Reserved image space to prevent layout shifts and used **caching** and **image optimization** (enforced webp / avif formats and compression to < **100 KB**) to reduce LCP: **FCP < 0.3s, LCP < 0.5s, CLS < 0.001**.
- Treated keyboard-accessible navigation, accessible form-error handling, semantic structure, and Lighthouse / axe checks as release requirements rather than post-launch fixes.
- Built a Strapi plugin for AI content generation with AI SDK.

React Native Skia App

2024

- Owned the photobooth app from design and development through release on the Apple App Store and Google Play.
- Users take a selfie, load it onto the canvas, edit with stickers and pen tools, then upload and share.
- Built the canvas with React Native Skia, React Native Reanimated, and gesture handlers; captured images with react-native-vision-camera.
- Converted SVG stickers into Skia SVG React components, enabling resize, rotate, pan, and scale on the canvas.
- Managed state with **Zustand**; kept selections atomic and debounced high-frequency updates.
- Optimized with memoization and context isolation; held **55+ FPS** in RN Perf Monitor with over 100 stickers and strokes on the canvas.

React Native Student App

2023

- Owned student flows (course catalog, booking, announcements) from Figma through release on the Apple App Store and Google Play.
- Maintained **99%** crash-free sessions after launch by diagnosing crashes, device compatibility issues, and API bottlenecks with **Sentry**.
- Data fetched with **Tanstack Query**, cut duplicate fetches with cache and staleTime; client state managed with **Redux Toolkit**.
- Reduced feed scroll jank after profiling identified JS-thread stalls from long lists and full-resolution images; migrated lists to **FlashList** with row memoization, cached and downscaled feed images, and ran **Reanimated** transitions and gesture handlers on the native thread.
- Froze inactive tabs and blurred screens, and deferred mounts, to prevent unnecessary re-renders.
- Routed promo and milestone notifications to the matching screen via Firebase Cloud Messaging deep links.

OTHER WORK

Also shipped restaurant loyalty/membership apps (React Native), a homebound elderly care app (Flutter), campaign / e-commerce storefronts (Next.js, Shopify **GraphQL**), CMS content sites, and multiple RBAC admin systems; Backend APIs (Node.js, Hono, Drizzle, Postgres)

SKILLS

Languages: TypeScript, Python

Frontend: React, Next.js, React Native, Tailwind CSS, shadcn/ui, Figma, Zustand, Redux, TipTap

UI behavior: TanStack Query, TanStack Form, React Hook Form, XState, TanStack Table, next-intl

Architecture: state modeling (URL / server / local), design systems, tokens, React composition, component APIs, routing, SSR/ISR, GraphQL, tRPC, OpenAPI, Storybook, Fumadocs

Backend: Node.js, Hono, NestJS, Drizzle, Prisma, OpenAPI, tRPC, Postgres, DynamoDB, Better Auth, Redis, websockets

Auth & security: OAuth 2.0 / OIDC, JWT, HttpOnly cookies, CORS, RBAC

AI: Mastra, Vercel AI SDK, LangChain, Pinecone, Docling, Strands, AWS Bedrock, Mem0, Langfuse

Infrastructure & DevOps: SST, Terraform, Docker, GitHub Actions, AWS, Vercel, Turborepo, Kafka, microservices (familiar)

QA: Vitest, React Testing Library, Playwright, Cypress, Lighthouse, axe, Sentry

Mobile: Expo, FlashList, Reanimated, Moti, Skia, Unistyles, FCM

QUALIFICATIONS

- AWS Certified Solutions Architect – Associate, 2026
- AWS Certified Cloud Practitioner, 2025

EDUCATION

University College London

2016 – 2020

- MPharm Pharmacy (undergraduate) - First Class Honours
- Transitioned from pharmacy to software development after finding passion in building applications.